

Title:

Matrix operations Tools using NumPy.

Objective:

To Perform various matrix operations through a Python-based interactive tool.

Operations Implemented

- Addition
- Subtraction
- Multiplication
- Transpose
- Determinant

Tools Used:

- Python
- NumPy

Results:

```
===== MATRIX OPERATIONS TOOL =====
1. Matrix Addition
2. Matrix Subtraction
3. Matrix Multiplication
4. Matrix Transpose
5. Matrix Determinant
6. Exit
Enter your choice (1-6): 1

Enter number of rows for Matrix A: 2
Enter number of columns for Matrix A: 2
Enter elements of Matrix A row-wise:
Row 1: 5 3
Row 2: 4 6

Enter number of rows for Matrix B: 2
Enter number of columns for Matrix B: 2
Enter elements of Matrix B row-wise:
Row 1: 8 4
Row 2: 5 2

Result (A + B)
-----
[[13.  7.]
 [ 9.  8.]]
-----
```

===== MATRIX OPERATIONS TOOL =====

1. Matrix Addition
2. Matrix Subtraction
3. Matrix Multiplication
4. Matrix Transpose
5. Matrix Determinant
6. Exit

Enter your choice (1-6): 2

Enter number of rows for Matrix A: 2

Enter number of columns for Matrix A: 2

Enter elements of Matrix A row-wise:

Row 1: 5 8

Row 2: 3 6

Enter number of rows for Matrix B: 2

Enter number of columns for Matrix B: 2

Enter elements of Matrix B row-wise:

Row 1: 5 3

Row 2: 4 2

Result (A - B)

[[0. 5.]

[-1. 4.]]

===== MATRIX OPERATIONS TOOL =====

1. Matrix Addition
2. Matrix Subtraction
3. Matrix Multiplication
4. Matrix Transpose
5. Matrix Determinant
6. Exit

Enter your choice (1-6): 3

Enter number of rows for Matrix A: 2
Enter number of columns for Matrix A: 2
Enter elements of Matrix A row-wise:
Row 1: 3 8
Row 2: 4 12

Enter number of rows for Matrix B: 2
Enter number of columns for Matrix B: 2
Enter elements of Matrix B row-wise:
Row 1: 4 2
Row 2: 3 9

Result ($A \times B$)

[[36. 78.]
 [52. 116.]]

```
===== MATRIX OPERATIONS TOOL =====
```

1. Matrix Addition
2. Matrix Subtraction
3. Matrix Multiplication
4. Matrix Transpose
5. Matrix Determinant
6. Exit

Enter your choice (1-6): 4

Enter number of rows for Matrix: 2

Enter number of columns for Matrix: 2

Enter elements of Matrix row-wise:

Row 1: 1 2

Row 2: 4 5

Transpose

[[1. 4.]

[2. 5.]]

===== MATRIX OPERATIONS TOOL =====

1. Matrix Addition
2. Matrix Subtraction
3. Matrix Multiplication
4. Matrix Transpose
5. Matrix Determinant
6. Exit

Enter your choice (1-6): 5

Enter number of rows for Matrix: 2

Enter number of columns for Matrix: 2

Enter elements of Matrix row-wise:

Row 1: 3 5

Row 2: 21 5

Determinant = -90.0